

Rubric - Merkle Trees Part

Student 1: _____
 Student 2: _____
 Student 3: _____

Section: _____
 Section: _____
 Section: _____

Criteria	Marks	Obtained
Data Preprocessing — Loads at least 1,000,000 records; correct parsing, cleaning, normalization; generates unique IDs when missing.	10	
Merkle Tree Construction — Merkle built from-scratch using SHA-256; correct leaf hashing, parent-node computation and final Merkle Root.	25	
Existence Proofs & Verification — Returns correct proof paths for review/product IDs; proofs verify correctly; proof verification ≤ 100 ms.	10	
Tamper Detection — Detects modification/deletion/insertion; reports mismatched hashes/altered nodes; simulation of tampering works.	10	
Performance Measurement & Metrics — Measures hashing speed, proof latency, total build time, peak memory; data presented and analysed.	15	
CLI Interface & Usability — Tabular dataset display with unique IDs; menu options (build, save/compare root, gen proofs, simulate tamper, run tests) are implemented and usable.	5	
Test Suite (correctness & perf) — Unit tests for correctness + performance tests (including ≥ 1 M run) and reproducible commands/results.	15	
Documentation & Final Reports — Performance report + analytical report: methods, complexity, results, interpretation, limitations & future work.	10	
Total	100	
Viva		

Bonus (GUI) — Visual tree, animated proof paths, interactive dashboard
Viva will have a significant weightage of the project so come prepared.